

SN Pointing Analysis - Summary Report

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Overall Summary

Total Categories Analyzed: 3

Scenarios Completed: 3 / 3

Best Case (True Electron Directions)

Angular Resolution:

Median: 4.99° ± 0.16°

68th percentile: 5.09°

90th percentile: 5.21°

Cluster Statistics (per category average):

Input clusters: 20714.0 (main: 4121.7, non-main: 16477.0)

Matched across 3 planes: 38.0 (main: 41.3, non-main: 0.0)

Matching efficiency: 0.18%

ED Network (Perfect MT/CT Tagging)

Angular Resolution:

Median: 6.99° ± 0.31°

68th percentile: 7.21°

90th percentile: 7.47°

Cluster Statistics (per category average):

Input clusters: 19990.3 (main: 3906.0, non-main: 16273.0)

Matched across 3 planes: 38.0 (main: 41.7, non-main: 0.0)

Matching efficiency: 0.19%

Full Pipeline (MT → CT → ED)

Angular Resolution:

Median: 10.21° ± 0.32°

68th percentile: 10.22°

90th percentile: 10.23°

Cluster Statistics (per category average):

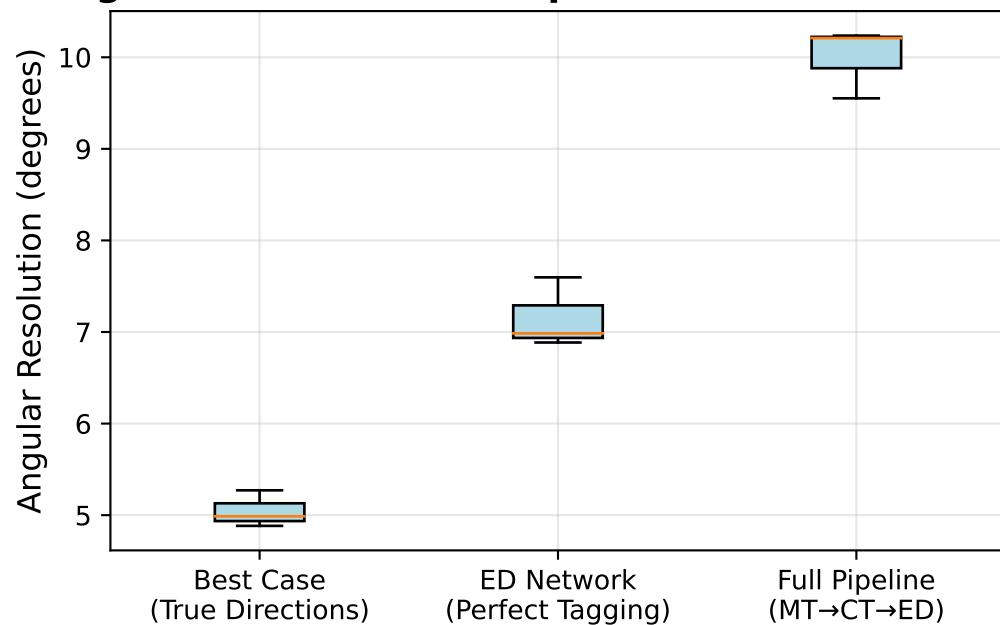
Input clusters: 19876.7 (main: 3828.0, non-main: 16077.7)

Matched across 3 planes: 40.7 (main: 42.7, non-main: 0.0)

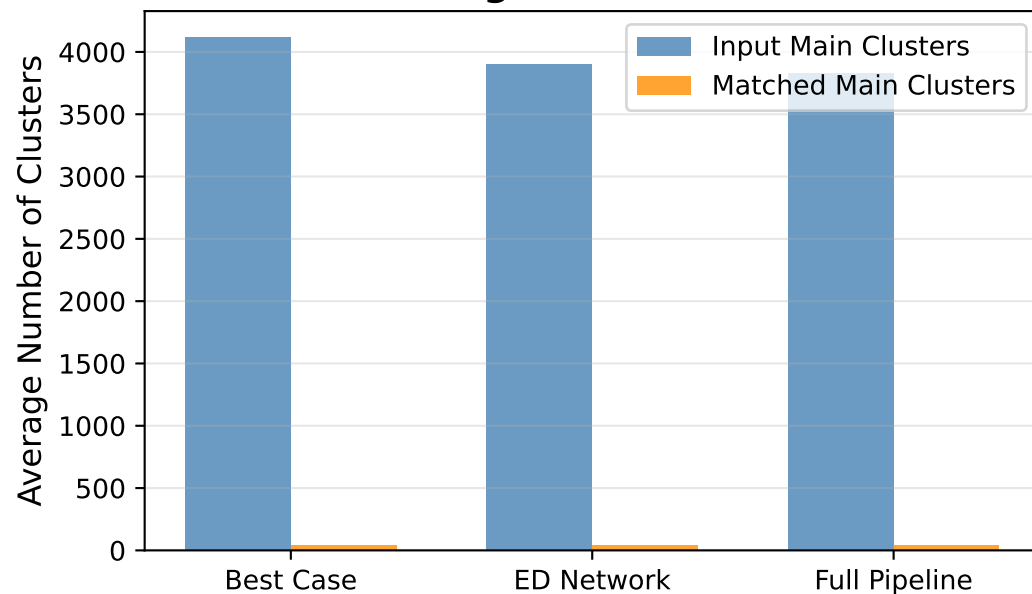
Matching efficiency: 0.20%

SN Pointing Analysis - Detailed Statistics

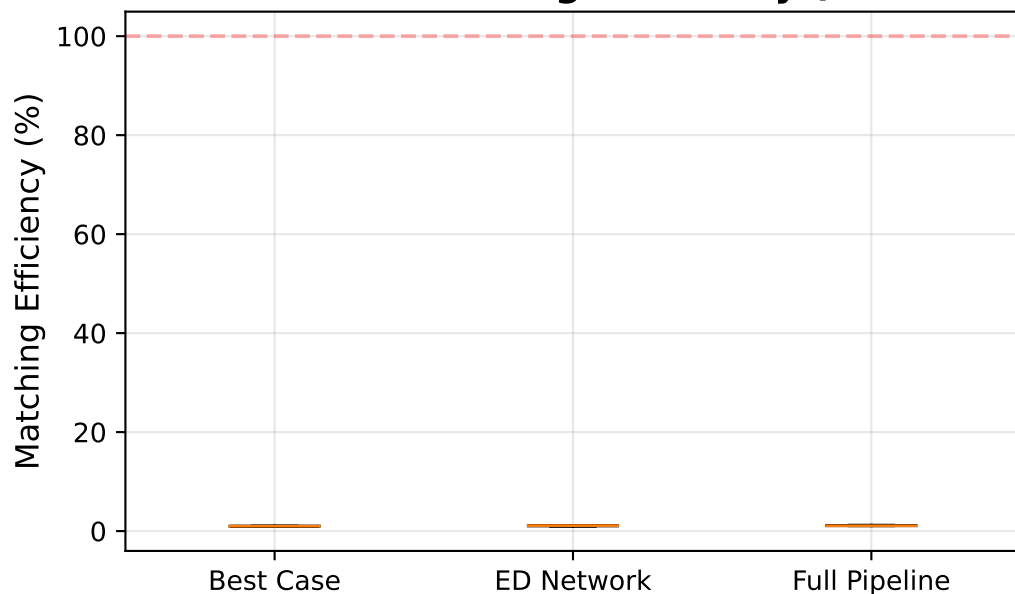
Angular Resolution Comparison Across Scenarios



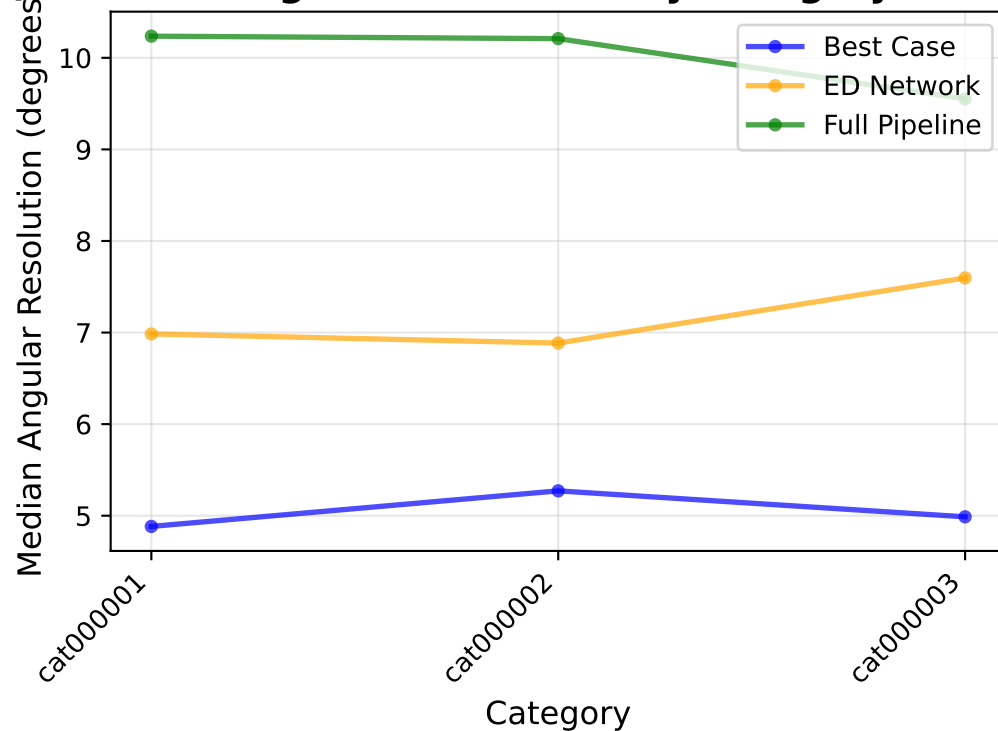
Cluster Filtering Statistics (Main Tracks)



3-Plane Cluster Matching Efficiency (Main Tracks)



Angular Resolution by Category



SN Pointing Analysis - Cosine Distribution (68% Quantile)

Cosine Distribution of Angular Errors

